

Counterparts

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Comments welcome

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Chapter 2

Arguments about A- and B-theories

The first six sections of this chapter will discuss an array of arguments for and against the different answers to questions T1, T2, M1, and M2 from chapter 1. The final section introduces counterpart theory, and explores its relation to some of these arguments.

2.1 Explicit quantification over times and worlds in semantics

Languages allow us to construct sentences by combining smaller linguistic expressions. Sometimes, some of the elements that combine to make a sentence are themselves sentences. From the point of view of semantics, these cases of sentences embedding sentences fall into two different kinds. In the first kind of case, exemplified by

- (1) Snow is white and grass is green

all that matters about the embedded sentences, as regards determining what proposition will be expressed by the outer sentence, is which propositions *they* express. Let us say in such cases that the embedded sentences occupy ‘propositional contexts’. In the simple case where all the sentences are context-insensitive (express exactly one proposition), this means that we can replace one of the inner sentences with any other context-insensitive sentence expressing the same proposition, without affecting which proposition is expressed by the outer sentence. To generalise this idea to context-sensitive sentences, we could work in a picture where there are many “admissible interpretations” of the language, each of which determines a mapping from sentences to unique propositions; an embedded sentence occupies a propositional context just in case, on any admissible interpretation, replacing it with another sentence that expresses that proposition *on that interpretation*

makes no difference to the proposition expressed by the outer sentence *on that interpretation*.¹

Paradigm *non*-propositional contexts are those involving quantification, for example

(2) Every farmer is surprised that his bean-plants are so tall.

On the most natural way of hearing this sentence, there seems to be no useful way to make sense of questions like ‘What proposition does “his bean-plants are so tall” contribute to the proposition expressed by (2)?’, or ‘What proposition was expressed by “his bean-plants are so tall” on this particular occasion?’ When an embedded sentence contains pronouns bound from outside, we need some different, and more flexible, way of talking about its role in determining what is expressed by the outer sentence. The standard way of doing this is derived from Tarski: it involves characterising the semantic profile of a sentence as a function from *assignment functions* to propositions, in such a way that the proposition expressed by a sentence on one assignment function can depend on facts about the propositions expressed by constituent sentences on other assignment functions. This approach requires the rather artificial expedient of assigning numerical “indices” (or something similar) to occurrences of pronouns and pronoun-binders. But the fact that it classifies the relevant contexts as non-propositional is no mere artefact; this is a feature we should expect any reasonable treatment of quantification to preserve.²

The contrast between propositional and non-propositional contexts need not be limited to cases where one sentence literally contains another one. It can also be drawn for any systematic syntactic mechanism that takes one sentence (perhaps subject to certain restrictions) and yields another. For example, as well as the embedding operation that turns a sentence *S* into the sentence ‘It is not the case that *S*’, we can consider the less artificial operation that applies to sentences that have a main verb, and involves negating that verb: this operation takes ‘Snow is white’ to ‘Snow is not white’, ‘Brutus stabbed Caesar’ to ‘Brutus didn’t stab Caesar’, etc. We might speculate that at the level of syntactic representation that matters for compositional semantics, the negated sentences do literally contain the sentences of which they are negations. But whether or not this is so, we

¹We might or might not want to go on to explain the notion of an “admissible interpretations” in terms of the Kaplanian ideology of “expressing relative to a context”.

²**Cite: Yli-Vakkuri and Rabern**

can consider the operation in question, and ask whether we need to know anything beyond the proposition expressed by its input sentence in order to settle the question what the proposition is to be expressed by its output sentence. In this particular case, the answer seems to be 'no'.

As we saw in section 1.2, there is good reason for propositional eternalists to classify contexts like 'always, ...' and 'it was once the case that...' as non-propositional. Likewise, there is good reason for propositional necessitarians to classify contexts like 'necessarily, ...' and 'it is contingent whether ...' as non-propositional. Because of this, it is natural for the eternalist or necessitarian to look for semantic accounts of temporal or modal embedding that appeal to the same formal apparatus standardly used for ordinary quantification. In the present section, we will see that there are strong arguments for the non-propositionality of these contexts, and for parallels between their semantics and that of ordinary quantification, which do not require propositional eternalism or necessitarianism as premises. These arguments tend to undermine a certain formerly-popular research programme in semantics, which was based on the idea that certain syntactic constituents in natural languages, or operations for transforming certain sentences of natural languages into others, should be treated in the same way as the temporal and modal operators that occur in the artificial languages studied by temporal and modal logic. We will have to see whether the arguments do anything to undermine propositional temporalism or contingentism themselves.

Consider (3):

- (3) Mary was in the same class as a girl whose mother was in college with someone who was working for Clinton.

This sentence has many structural disambiguations, but only some of them will concern us. Let us forget about the readings on which the sentence requires that at some past time, Mary was in the same class as a girl whose mother was in college at that very time. The remaining readings all require there to be two past times, t_1 and t_2 , such that at t_2 , Mary is in the same class as a girl whose mother is in college at t_1 . One reading—call it reading A—requires, in addition, that someone who is in college with the mother at t_1 is working for Clinton at t_2 . Another reading—reading B—requires that someone who is in college with the mother at t_1 is working for Clinton at t_1 (presumably as a part-time job). There are still further readings that allow the work for Clinton to take place at some third time t_3 ; this time t_3 can be required either to be earlier than t_1 , earlier than t_2 , or merely earlier than

the present. But let us focus on readings A and B. What kind of structural ambiguity is this? It has a familiar feeling: it looks just like the kind of structural ambiguity that arises when a pronoun occurs in the scope of more than pronoun-binding constituent (such as a quantifier), and can thus be read as bound in several different ways:

- (4) Every miller knows at least one farmer who throws stones at his donkey

Here 'his' can be bound by 'every miller' or by 'at least one farmer'. Most current work in the semantics of tense follows this analogy where it leads. In (3), the relative clause 'who was working for Clinton' is taken to contain a *free variable* ranging over times, and the material earlier in (3) is taken to include two different quantifiers over times, each capable of binding the free variable in the relative clause. There is some debate as regards which syntactic constituents of (3) are the variables and which are the quantifiers.³ But what matters for our point of view about this dominant approach (one that posits 'explicit quantification over times in the object language') is that *all* sentences involving tense morphemes—even simple ones like 'John was working for Clinton'—need to be assigned the kind of semantic profile appropriate to sentences containing bindable elements, such as 'He throws stones at his donkey'. For any sentence involving tense morphemes can be embedded under further tenses (or under 'will', which is not strictly speaking a tense), and will give rise to different readings depending on how we set up the binding relations. And the crucial thing about this kind of profile is that it cannot be captured by associating them once and for all with a single proposition: we will need something more complicated, like a function from assignment functions to propositions.

The time variable in an embedded tensed expression is sometimes bound by some quantifier higher in the same sentence. But not all of the time variables are bound in this way, since the quantifier that binds one variable will introduce a new free variable of its own—these quantifiers work like restricted quantifiers of the form ' $\exists t_1 : t_1 < t_0 \dots$ '. What happens to the free variables that are not bound? A standard thing to say about utterances of sentences involving free variables is that 'the context provides' an assignment function which maps these variables to entities of an appropriate kind; the proposition expressed by the speech act is the one that is expressed

³For example, Kusumoto (1999) favours a system in which the tense morphemes are the variables and the quantifiers are phonologically null. Others **cites!** assign quantifier-like semantic profiles to some tense morphemes.

by the sentence relative to this contextually-provided assignment function. Thus, for example, if I utter 'He is a fool', there will be some particular object x which is assigned to the pronoun I uttered by the context I am in, and the proposition I express or assert is the proposition that x is a fool. If we apply this standard approach to the free time variables posited by the dominant view, we will end up with a view according to which tensed sentences always express eternal propositions. For example, if I just say 'John was working for Clinton', I express a proposition concerning some particular time t . On one approach, t is a particular time earlier than the time at which I am speaking, and what I say is that John is working for Clinton at t . On another approach, t is the time of my speech, and what I say is that John is working for Clinton at some time that is earlier than t (and perhaps meets certain further conditions). Either way, what is expressed is an eternal proposition. Thus, the conjunction of the standard approach to free variables with the semantic theory that posits such variables leads to the view that all propositions semantically expressed by sentences in natural languages are eternal propositions.⁴

If this is a good argument, there is a comparably good argument for the claim that all propositions semantically expressed by sentences in natural languages are non-contingent. For the linguistic phenomena associated with mood are strongly analogous to those associated with tense. For example, consider a modal version of (3):

- (5) Mary would have been in the same class as a girl whose mother would have been in college with someone who would have been working for Clinton.

This has a range of disambiguations exactly analogous to those of (3). If we take the counterfactual supposition relevant for the first 'would' to be different from the counterfactual supposition relevant for the second 'would', we have the option of interpreting the third 'would' as referring us back either to the first supposition or to the second, or of introducing some further supposition. Again, there is a formal parallel with the structural ambiguities associated with variable-binding; and most contemporary work in the semantics of modals takes this very seriously, by positing an apparatus of free variables ranging over worlds, and quantifiers binding such variables, in the syntax of the object language, even in very simple sentences. (See e.g.

⁴King 2003 is an influential defence of propositional eternalism in which a crucial role is played by the success of semantic theories positing explicit quantification over times.

Stone 1997, Percus 2000, Schlenker 2006, Speas 2004, p. .) If we plug in the standard treatment of free variables, this will result in a picture whereby the proposition we express when uttering a sentence, say ‘Snow is white’, is a proposition about a particular world, to the effect that snow is white *at that world*, and thus a necessary proposition. Schaffer (2012) gives exactly this argument for the claim that all the propositions expressible in natural language are necessarily true or false.

Of course, it is far from obvious that all the propositions that there are are semantically expressible in natural languages. Thus, in principle, a propositional temporalist could just accept the conclusion that all propositions semantically expressible in natural languages are eternal, and a propositional contingentist could accept the conclusion that all propositions semantically expressible in natural languages are non-contingent. However, there are at least two reasons to doubt that these positions are stable.

First reason: anyone who accepts the above argument for the claim that every proposition semantically expressible by a sentence in natural language is eternal [non-contingent] should also accept the claim that every proposition semantically expressed *relative to some assignment* by a sentence in natural language is eternal [non-contingent]. For the argument for the claim about the unrelativised expression relation turns on combining that claim about the assignment-relative expression relation with the standard treatment of free variables as getting assigned values by the context.⁵ But the latter claim looks very problematic for a propositional temporalist [contingentist]. For consider the following instances of the *Temporal Truth Schema* and the *Modal Truth Schema*:

- (6) a. For any proposition p , the proposition that p is true is such that always, it is true iff p is true.

⁵Even setting the above argument aside, the combination of the claim that no proposition expressible *simpliciter* in a natural language is temporary [contingent] with the claim that some propositions expressed by sentences in natural languages relative to certain assignments are temporary [contingent] looks very unpromising. For suppose ‘ $\phi(x_1, \dots, x_n)$ ’ expresses a certain temporary or contingent truth p relative to a certain assignment g . Then what could stop us from introducing directly referential names $a_1, \dots, a_n$ for $g(x_1) \dots$ and $g(x_n)$, and thereby seeing to it that $\phi(a_1, \dots, a_n)$ will express p *simpliciter*? One might, I suppose, think that the only relevant entities that enter into temporarily [contingently] true propositions in this way are too similar to one another (intrinsically and in their relations to us) for us ever to succeed in naming them. But the idea that there are such limits on our capacity for demonstration looks unpromising: the fact that each material object is extremely similar to many others does not prevent us from expressing propositions about particular material objects.

- b. For any proposition p , the proposition that p is true is such that necessarily, it is true iff p is true.

These claims look are overwhelmingly plausible: even if we reject the (quite attractive!) view that each proposition is *identical* to the proposition that it is true, there is clearly an intimate enough relation between p and the proposition that p is true to make the idea that they could differ in respect of being eternal or contingent utterly bizarre. Moreover, as we saw in section 1.2, these claims very plausibly follow from the *Truth Schema*, which we are holding fixed as a key part of the proposition role. But anyone who accepts (6a) [(6b)] and holds that there is a temporarily [contingently] true proposition must also accept the following sentence: ‘There is a proposition p such that the proposition that p is true is temporarily [contingently] true’. But for this sentence to express a truth, there must be some variable assignment under which ‘the proposition that p is true’ denotes a temporarily [contingently] true proposition. But it is hard to imagine a sensible view on which the proposition denoted by ‘the proposition that ϕ ’ on a given assignment could be distinct from the proposition expressed by ϕ on that assignment.

Second reason: the view that according to which there are temporarily [contingently] true propositions, but no ordinary human being ever *entertains* or *believes* any of them looks extremely unattractive. For one thing, it looks hard to motivated: why would one posit such propositions, if one did not regard them as potential objects for attitudes like belief? For another thing, it leads to a bizarre skepticism when combined with temporal [modal] elitism: one instant [world] has an especially natural unit class, but we can never know anything about which instant it is. Indeed we cannot even *think* the answers to this question, since any attempt to do so will merely lead to our thinking something eternally [necessarily] true or false. But if ordinary humans sometimes *believe* transient [contingent] propositions, we can hardly deny that they also sometimes *assert* them. If we are capable of believing them, we will certainly sometimes also act in such a way as to get other people to believe them. And while asserting a proposition isn’t just doing something which you intend to induce belief in that proposition, it would be implausible to think that whatever further conditions need to be in place for one to count as asserting a proposition are never satisfied in the case of transient [contingent] propositions. Once we get this far, it looks hard to resist the thought that certain natural-language sentences are regularly, and as a matter of linguistic convention, used to assert transient [contingent] propositions. And if that is the case, on what grounds could one deny

that those sentences are capable of *semantically expressing* such propositions? Those who resist at this point will be vulnerable to the accusation that, by allowing the question which propositions a sentence can semantically express to float free from considerations about what that sentence can be used to assert, they are turning the relation of semantic expression into something idle and useless. We do not need to be in possession of anything like a worked out metasemantic theory to see that there needs to be some close connection between the semantic values of sentences and what they are used to assert.

It is thus hard to sustain the view that there are temporarily [contingently] true propositions, but they play no role in natural-language semantics. Thus propositional temporalists and contingentists have good reason to want to resist the above arguments for the conclusion that natural-language sentences never semantically express such propositions. This illustrates the more general point that there is no hope of delimiting metaphysics from semantics in such a way that arguments based in natural-language semantics can be dismissed as irrelevant to metaphysical conclusions. This is a fact to be celebrated, since our inquiries in metaphysics are likely to go best when we can bring to bear as many external sources of discipline as possible.

So, how might temporalists and contingentists resist the argument from the success of semantic theories positing “explicit” time and world variables in natural-language syntax to the conclusion that all the propositions semantically expressed by sentences in natural languages are eternal or non-contingent? One strategy is simply to reject those semantic theories. They are not the only game in town: alternative ways to account for the relevant data (such as the ranges of readings available for (3) and (5)) are known. One hallmark of the alternative approaches is that the presence of bindable elements is an *optional* matter: they can pop up when they need to in an embedded sentence, but they can also be entirely absent. (See e.g. Cresswell 1990) Among linguists, the consensus among those who do not regard the issue as merely verbal seems to be that it is theoretically superior to posit ubiquitous explicit time and world variables. For example, it is argued that this will allow us to give a general and explanatory theory which unifies the familiar constraints on the allowable patterns of binding of pronouns with certain constraints to which the binding of time and world variables seems to be subject (Percus 2000, Stechow 2004). What should we make of arguments like these? On the one hand, it is hard to believe that the relevant explanatory advantages could really be significant enough to ground an ar-

gument for metaphysical theses as weighty as eternalism or necessitism. On the other hand, would *you* want to be the philosopher who tells the linguists that their elegant, unified formal treatment of individual, time, and world variables must be given up on the grounds that it is inconsistent with the truth of temporalism or contingentism?

Fortunately, there is no need to do anything so heroic: there are several ways to reconcile temporalism and contingentism with a semantic theory that posits ubiquitous time and world variables. One strategy is developed by Ninan (2010). Ninan proposes (on behalf of the temporalist) that the time variable in a simple sentence like ‘Obama is president’ is not really free: rather, it is bound by a special unvoiced element which is always inserted at the beginning of every sentence, and typically binds the time variables which would otherwise be free, or at least (or at least, those associated with present tense morphemes). One could resist the argument for necessitism in exactly the same way by positing an unvoiced binder that sweeps up world variables that have not yet been bound. Ninan’s temporalist is working in a framework where propositions are identified with functions from times and worlds to truth values, so the silent binder he posits takes the form of a lambda abstractor. At logical form, ‘Obama is president’ looks like $\lambda t(\text{President}(\text{Obama}, t))$: its semantic value is the function that maps each time i to the function that maps each world w to Truth if Harry is asleep at i at w and Falsehood otherwise. However, the basic strategy of positing an unvoiced binder could be adapted to other accounts of the nature of propositions. For instance, we might take the binder to be itself an existential quantifier restricted to the one and only present instant, giving the logical form $\exists t : \text{Present}(t)(\text{President}(\text{Obama}, t))$.

There is another approach which avoids the need to posit any such unvoiced binder. According to this approach, the constituents we have been calling ‘time variables’ and ‘world variables’ do not behave in all respects like variables as standardly conceived. They can be bound, and when they are, their semantic contribution is just like that ordinary variables. But when they are not bound, they need not function as devices for referring directly to times or worlds; instead, they can work like *nonrigid definite descriptions*. For example, if ‘Obama is president’ looks at LF like $\text{President}(\text{Obama}, t_0, w_0)$, ‘ t_0 ’ and ‘ w_0 ’ could be semantically equivalent to the nonrigid descriptions ‘the present time’ and ‘the actualised world’, making the whole sentence express the transient, contingent proposition that Obama is president at whichever time is present, at whichever world is actualised. This is of course

equivalent to the proposition that Obama is president, since it is always and necessarily true that whatever is the case is the case at the present time at the actualised world.

Neither of these defensive moves is completely cost-free. Eternalists and necessitists can argue against Ninan's theory on the grounds that he needs to posit an unvoiced element (the hidden lambda abstractor) for which there is no independent evidence. And they can argue against the nonrigid-description theory on the grounds that it needs a simpler and less elegant account of the semantic roles of the relevant variable-like element than the one provided by the standard account.

But neither of these objections is very strong. First, as regards Ninan's theory, it is true that in general the positing of unvoiced elements needs to be justified on syntactic grounds, but in semantic theories using variables, it is quite common to make an exception to this for lambda-abstractors—certainly the abstractors that occur in the syntactic representations that serve as inputs to semantics in the system of Heim and Kratzer (1998) are not justified by any arguments that look to have a distinctively syntactic flavour. Semanticists seem comfortable with a relatively instrumentalistic attitude towards the lambda abstractors that occur in their LF representations—they seem to accept the positing of such elements on the distinctively semantic grounds that they are needed so as to avoid crashes due to type mismatch. Against this background, it is not clear that anyone is in a position to reject Ninan's view on the grounds of syntactic complexity.

Second, as regards the nonrigid-description theory, the positing of constituents that can sometimes behave like bound variables and sometimes like nonrigid descriptions would indeed be a cost if such creatures were unknown elsewhere in syntax. But this is far from being the case: in fact *pronouns* seem capable of behaving both like bound variables and like nonrigid descriptions. And since the most compelling arguments for positing ubiquitous world and time variables come from analogies between binding-like phenomena that arise in connection with tense and mood and the binding of pronouns (Partee 1973), there is nothing in any way unprincipled about an explanation of these phenomena that appeals to items that share pronouns' capacity to function, when unbound, like nonrigid descriptions.

The best-known cases where pronouns are arguably neither bound nor directly referential are in sentences like 'Every farmer who owns a donkey beats it'. Now is not the right place for a survey of the formidable formal machineries that have been devised to account for the behaviour of these

“donkey pronouns”. But the point we need can be made without delving into these complexities. A disappointed gameshow contestant who only won a toaster might explain her disappointment as follows:

- (7) If only I had been able to remember who directed ‘The Maltese Falcon’, it would have been a car

This obviously does not require that the toaster I won is something which could have been a car; the pronoun ‘it’ seems in context to work to work like the nonrigid definite description ‘the prize’. ⁶

Moreover, there is independent evidence that might suggest that the variable-like elements associated with tenses and moods can behave like non-rigid descriptions. Consider

- (8) I didn’t mention the war

(8) can (*pace* Bach 1994) express a true proposition even if the speaker has mentioned the war at some past time or other. One influential explanation of this fact, suggested by Partee (1973), is that the past tense morpheme in ‘didn’t’ deictically refers to a certain temporal interval, and the proposition expressed by the sentence is true just in case that particular interval is one during which I don’t mention the war. But this treatment of the relevant form of context-sensitivity seems unduly inflexible. Surely I could use (8) to express the proposition that I didn’t mention the war *during a certain dinner party*, where the dinner party in question could have taken place earlier or later. We can confirm that such non-rigid readings are possible by considering embedded contexts:

- (9) I managed not to mention the war, but only because they arrived so late that I was half asleep the whole time.

There may have been no risk that I would mention the war at that particular interval, since if I hadn’t been half asleep during it, I would have been all the way asleep. What was averted by the lateness of the arrival was not a mentioning of the war during any particular interval, but rather a mentioning of the war during *that* dinner (or to *those* guests). One way to explain this which stays relatively close to Partee is to say that the past

^{6**}Discuss Geach on ‘Pronouns of laziness’ and look at his examples.**

tense morpheme works in this case like the nonrigid definite description 'the interval occupied by that dinner party'.^{7 8}

Thus, the analogies between tenses and moods and pronouns do not provide strong grounds for rejecting temporalism or contingentism. For the analogies are indeed quite far-reaching, and include phenomena that seem to parallel those occurrences of pronouns that function like nonrigid descriptions. So, temporalists and contingentists will not need to posit any novel syntax or semantics in order to claim that the relevant nonrigid descriptions are often ones like 'the present instant' and 'the actualised worlds', which denote times and worlds in virtue of their transient or contingent A-theoretic properties.

⁷Alternatively, we could say that the past tense morpheme functions as a *bound* variable, where the binder is an unvoiced quantifier restricted to times that are earlier than the particular instant at which the sentence is uttered, and at which the dinner party is going on.

⁸**Discuss Ludlow's approach**